The Survival Function

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Introduction

For a time-to-event random variable $T \geq 0$, the **survival function** $S(t) = P(T > t)$ is the probability of not having experienced the event by time t. It is the complement of the CDF and the single most important descriptive quantity in survival analysis.

Prerequisites

Random variables, CDF, basic survival-analysis concepts.

Theory

For a non-negative random variable T,

$$S(t) = P(T > t) = 1 - F(t).$$

Properties:

- $S(0) = 1$, $\lim_{t \to \infty} S(t) = 0$.
- Non-increasing.
- Right-continuous (by convention).

Related quantities:

- PDF: $f(t) = -S'(t)$.
- Hazard: $h(t) = f(t)/S(t)$.
- Cumulative hazard: $H(t) = -\log S(t)$.

Classical forms:

- Exponential: $S(t) = e^{-\lambda t}$.
- Weibull: $S(t) = \exp\big[-(t/\lambda)^k\big]$.
- Log-normal: $S(t) = 1 - \Phi\big((\log t - \mu)/\sigma\big)$.

Kaplan-Meier is the non-parametric MLE of S from right-censored data: a step function that drops at each observed event time by a factor reflecting the fraction of the at-risk set that experiences the event.

Assumptions

Non-negative random variable representing a time to an event. For Kaplan-Meier: independent censoring.

R Implementation

```r
library(survival)
set.seed(2026)

# Theoretical survival functions
t_grid <- seq(0, 10, length.out = 400)
```

```r
S_exp     <- exp(-0.3 * t_grid)
S_weibull <- exp(-(t_grid / 3)^2)
S_lnorm   <- 1 - plnorm(t_grid, meanlog = log(3), sdlog = 0.5)

plot(t_grid, S_exp, type = "l", col = "#6A4C93", lwd = 2,
     main = "Theoretical survival functions", xlab = "t", ylab = "S(t)")
lines(t_grid, S_weibull, col = "#2A9D8F", lwd = 2)
lines(t_grid, S_lnorm,   col = "#F4A261", lwd = 2)
legend("topright", c("exponential", "Weibull k=2", "log-normal"),
       col = c("#6A4C93", "#2A9D8F", "#F4A261"), lwd = 2)

# Empirical Kaplan-Meier from censored data
n <- 80
T_true <- rweibull(n, shape = 2, scale = 3)
C      <- runif(n, 0, 10)
T_obs  <- pmin(T_true, C)
delta  <- as.integer(T_true <= C)

km <- survfit(Surv(T_obs, delta) ~ 1)
plot(km, conf.int = TRUE, main = "Kaplan-Meier estimate",
     xlab = "t", ylab = "S(t)", col = "#2A9D8F")
lines(t_grid, exp(-(t_grid / 3)^2), col = "#F4A261", lwd = 2)

# Median survival
summary(km)$table["median"]
qweibull(0.5, shape = 2, scale = 3)
```

Output & Results

```
median
 2.499
```

```
[1] 2.498
```

Empirical and theoretical median survival times agree. The Kaplan-Meier step function closely tracks the true Weibull survival curve.

Interpretation

Reporting a median survival with its 95 % CI from a Kaplan-Meier estimate is the standard headline statistic of survival studies. Survival functions are typically compared across groups via log-rank tests and visualised together on the same axes.

Practical Tips

- Report medians with CIs when survival drops below 0.5; report specific-time survival (1-year, 5-year) when the study ends before the median is reached.
- Kaplan-Meier is distribution-free; parametric survival models (Weibull, log-normal) give smooth curves but commit to a shape.
- Include a risk table below Kaplan-Meier plots (`survminer::ggsurvplot(..., risk.table = TRUE)`).
- Under heavy censoring, the tail of the survival curve is poorly estimated; avoid over-interpreting late-follow-up differences.
- Survival functions never take values above 1 or below 0; if your empirical estimator returns such values, there is a bug.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Probability, conditional probability, Bayes' theorem
- Discrete distributions: Bernoulli, binomial, Poisson
- Continuous distributions and Q-Q plots